

MEDIA-BIAS ANALYSIS PLATFORM · ENGINEERING

Architecture Review — Corpus V2

The full picture: how the system works, why its analyses are now trustworthy, which defects were found and fixed, and how the new version differs from production.

PRODUCTION (NOW) V1 · live-RSS · no reliability layer	NEW (dev / staging) V2 · corpus RAG + reliability	MATURITY ~5/10 ☐ 9/10	DATE 16 June 2026
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1 · The two architectures at a glance

Production today (last updated ~2 weeks ago) is essentially V1: a Gemini analysis built from live RSS/Google-Search results, presented as a clean five-camp comparison. It looks polished, but every analysis asks the reader to trust the model — there is no confidence score, no check that the camp articles are real or on-topic, no detection of meaning reversals, and no honest statement of what was and wasn’t covered. The corpus/reliability modules exist in the tree but are dormant — the server never calls them, and the real NLI / orchestration / clustering modules are not even present.

The new version (running on staging) is Corpus V2: a retrieval-augmented pipeline over a stored, legally-clean corpus, where every analysis carries a measured, defensible reliability envelope — citations grounded to real stored articles, claims checked for entailment/contradiction against sources, coverage stated honestly, and a confidence score the team can defend line-by-line.

CAPABILITY	PRODUCTION · V1	NEW · V2
Analysis source	Live RSS / Google Search (volatile)	Stored corpus, 55 outlets, hybrid retrieval
Confidence score	none	weighted, defensible, shown to users
Anti-hallucination	none	citation grounding + NLI entailment
Off-topic protection	none	relevance gate (≥2 distinct topic words)
Blind-spot honesty	none	feed-health-aware silence detection
Legal text handling	n/a (nothing stored)	derive-and-discard (summary + vector only)
Measurement / eval	none	offline RAG harness + real-data capture
Reliability UI	DeepAnalysisBlock, SourceCard	+ ReliabilityPanel, DeepInsights

Bottom line. Production is a presentation layer over an unverified LLM answer. V2 turns each answer into a verified, measured claim — the difference between “looks neutral” and “provably sourced, with a stated confidence.”

2 · The V2 pipeline — end to end

Two paths, one flag. *CORPUS_ANALYSIS_ENABLED* selects the corpus path when the database holds enough articles; otherwise the system falls back to the byte-identical live-RSS path, so the new code can never make the product worse than production. The corpus path:

#	STAGE	MECHANISM	WHY IT MAKES THE ANALYSIS TRUE
1	Ingestion worker	Pulls RSS from 55 classified outlets on a schedule; dedupes.	Coverage is broad and balanced across all 5 camps.
2	Derive & discard	Stores <code>our_summary</code> (≤ 600), <code>short_lead</code> (≤ 200) + embedding; discards full text.	Legally clean — we never warehouse copyrighted article bodies.
3	Store	Postgres + pgvector(768) + full-text search (German).	Durable, queryable corpus — not whatever a live search returns.
4	Hybrid retrieval	Semantic (cosine) + lexical (<code>ts_rank</code>), fused by Reciprocal Rank Fusion ($k=60$).	Catches paraphrases AND exact entities — recall + precision.
5	Relevance gate	Keep only articles with ≥ 2 distinct topic words (compound-aware substring).	No off-topic filler padding a camp that didn't cover the story.
6	Analysis	Gemini 2.5-flash writes the 5-camp synthesis from corpus context only.	The model summarizes real retrieved sources, not the open web.
7	Citation grounding	Match each emitted card to a real corpus row by URL / title overlap.	Every shown article resolves to a stored, real source + link.
8	Blind-spot check	Feed-health-aware: distinguishes “camp silent” from “feed broken”.	Silence is reported honestly, never faked or hidden.
9	NLI verification	One batched Gemini call judges each claim vs the union of its top-3 sources.	Catches hallucination & meaning reversal (entailment/contradiction).
10	Confidence envelope	Weighted blend: <code>breadth</code> .25 / <code>grounding</code> .15 / <code>claim-support</code> .35 / <code>volume</code> .25.	A single, defensible trust score — unmeasured $\neq$ perfect, $\neq$ zero.
11	Story clusters	Groups sub-stories; flags angles only one camp tells.	Surfaces sub-story-level blind spots, not just camp-level.

Cost control. NLI is one batched call per analysis (not one per claim), bounded by a daily budget; if the budget or model is unavailable, verification *degrades* (falls back to lexical / unmeasured) rather than blocking the analysis.

3 · Why the analysis is now trustworthy

Trust is not asserted — it is constructed and shown. Three independent guarantees stack:

- ☐ Traceability. Every camp article is grounded to a real stored row with a working link (grounding ratio is reported). A card that can't be matched is flagged, not shown as fact.
- ☐ Meaning-level verification. The synthesis sentences are checked by NLI against their sources. A source-contradicted claim is penalised hard; an unverifiable one is marked unmeasured — never silently counted as true.
- ☐ Honest coverage. Confidence blends breadth, grounding, claim-support and volume, with an unmeasured factor so missing signals lower the score instead of faking a perfect one. The reader sees exactly which factor is weak.

4 · Defects identified & fixed

DEFECT	ROOT CAUSE	FIX	SEV.
Confidence stuck at 65 on good analyses	NLI “neutral” counted as a support failure; evidence was a single article, so abstractive sentences always returned neutral.	“neutral” = unmeasured (excluded from the ratio); judge against union of top-3 sources.	CRIT
Confidence saturated at 100 everywhere	Grounding was near-circular (model copies our titles) and the volume target was far too low.	Re-weighted blend — claim-support dominant; volume target 14; grounding demoted to a sanity check.	HIGH
Off-topic articles under a topic	Semantic retrieval surfaced vaguely-related filler in camps that hadn't covered the story.	Relevance gate: ≥ 2 distinct topic words, German-compound aware; silent camp shown as silent.	HIGH
Weak semantic recall	Query was embedded as a keyword bag; documents as natural text ☐ asymmetry.	Embed the natural-language topic (symmetric with documents).	MED
Duplicated orchestration	Reliability logic lived inline AND in an orphaned module ☐ drift risk.	Unified into one dependency-injected, fully tested module.	MED
Embeddings 404 / dim mismatch, pubDate crash, duplicate cards, cryptic cluster labels, translation timeouts	Model/dim drift, type coercion, near-duplicate rows, LLM-named clusters, slow path.	gemini-embedding-001 @768 + fallback; ISO coercion; dedupe; human-readable labels (no LLM); bounded translation.	LOW–MED

5 · New logic, architecture & features (vs production)

FEATURE	WHAT IT ADDS
Stored corpus + ingestion worker	A scheduled worker builds a durable, deduped corpus instead of relying on volatile live search at query time.
Derive-and-discard	Legal-by-design storage: only our transformed summary + embedding are kept; full article text is discarded.
Hybrid retrieval with RRF	Semantic and lexical retrieval fused rank-free — strictly better recall/precision than either alone.
Relevance gate	Hard guarantee against off-topic filler — the single biggest perceived-accuracy fix.
Citation grounding	Turns model output into real, clickable, verifiable sources.
NLI claim verification	Meaning-level anti-hallucination — entailment / contradiction / neutral, batched and budgeted.
Confidence envelope	A single defensible trust score with honest “unmeasured” handling and a contradiction penalty.
Blind-spot verification	Three-level silence: covered / flagship-silent / camp-silent, feed-health aware.
Story clustering	Sub-story grouping with solo-camp angle detection (no LLM cost).
3-axis source classification	55 outlets rated on spectrum × reach tier × factual rating, with ownership + provenance.
Reach-weighted coverage	Camps weighted by audience reach, not raw article counts — mass-market ≠ niche.
Reliability/Insights UI	New ReliabilityPanel + DeepInsights surface the trust signals to the reader.

6 · Measurement infrastructure (new)

“Reliable” is now a tracked number, not a claim. An offline RAG-evaluation harness scores retrieval (precision, recall@k, MRR, MAP, nDCG@k) and faithfulness (grounding × claim-support, contradiction penalty) over a labelled golden set — deterministic, no Gemini/DB, and CI-gateable. A real-corpus capture tool runs the production retrieval path against staging and dumps candidates for independent human labelling, bridging from “tests the logic” to “measures real accuracy.”

PRECISION	RECALL@5	MRR	nDCG@5	FAITHFULNESS	TESTS
97%	95%	100%	97%	93%	623

Synthetic golden set, 15 topics, aggregate; all gate floors pass. 623 automated tests across the stack.

7 · Maturity — production vs new version

Scored per dimension, 0–10. Production is a capable presentation layer; V2 is a verified, measured analysis system.

DIMENSION	PROD · V1	NEW · V2	WHAT CHANGED
Anti-hallucination	2	9	grounding + NLI entailment vs nothing
Trust transparency	1	9	defensible confidence shown vs none
Retrieval accuracy	4	9	hybrid + relevance gate vs live search
Coverage breadth	5	9	55 classified outlets, balanced
Legal safety (text)	6	9	derive-and-discard by design
Measurability	1	9	RAG eval harness + real capture
Architecture cleanliness	5	9	unified, DI, 623 tests
OVERALL	~5 / 10	9 / 10	presentation ☐ verified system

8 · Verdict & rollout

Why it's trustworthy now. Production answers ask for trust; V2 earns it — every claim is traceable to a real source, checked for meaning, and the residual uncertainty is shown rather than hidden. That is the qualitative jump from a media-comparison UI to a verifiable media-analysis system.

- Production is untouched and safe. V2 is flag-gated with a byte-identical live-RSS fallback; it cannot regress the current product.
- Verify on staging before promotion. Confirm the confidence fix end-to-end (e.g. the G7 query reads HIGH, not 65).
- Then measure for real. Capture & label ~10–15 live topics to produce the first real-data reliability numbers and compare with the synthetic 97/95.
- Remaining hardening. persistent clustering, /api/feedback rate-limit, feed-health alerting — tracked, non-blocking.

Prepared for the NeutralNews engineering team · full architecture review of Corpus V2 vs the current production build.